

Additional Details on Project Work

Assuming that you have already chosen a project topic, you shall submit a project plan this week (**until 5/10/25**). The requirements for passing this module are explained below. Your grade depends on the success of your project and the complexity of the features you implement. Please take note of the following:

- If you need additional time to refine your project ideas, you can submit your plan by the end of next week (**until 12/10/25**). If you submit your project plan earlier, you will get feedback on how to amend your plans or proceed with developing your ideas, designing and implementing your project.
- After reviewing your project plan, you will receive approval to go ahead with your ideas or amendments you should make if your project idea is not feasible, out of scope or does not satisfy the minimum requirements.
- It is expected that there may be challenges during the project execution process. You can ask questions through the Discussion Forum (a thread will be created for the project only).
- After completing your work, test your code thoroughly before uploading. Write a report describing the process you passed through, the challenges you encountered and how you overcame them.

Project plan (max 2 pages, not including cover page)

1. Write the title of your project
2. Write names of group members, if you work in groups (encouraged)
3. Specify objective. What does your application do? Who can use it, for what purpose?
4. List the features (functions) of your application.
5. Describe your source of data.
6. List the tech stack (tools, frameworks, packages, etc.) you intend to use. In case new tools are needed at a later point in the project, it can be updated.

Note: There are no specific templates for writing your plan. You can use your own template or write it in free format, but the items mentioned above should be present in the same order given.

Project Report (3-5 pages, not including cover page)

1. Explain the implementation process
2. Describe the steps you followed, choice of development environment, tools and frameworks, data collection and preprocessing, the computational logic you applied

3. If you used generative AI tools for coding or elsewhere, clearly specify which part of your work is **AI-generated** and which part is **your own**.
4. If any of the features promised is missing or its implementation failed, clearly explain why your delivery did not meet your initial expectations.
5. If you implement additional features other than the originally promised, list them separately and explain what they do
6. Describe the success/failure of the project in your opinion. Explain which of the promised features are successfully implemented.
7. Describe what you learnt specifically from the project work.
8. Provide a short user guide (readme note) that anyone who wants to use your work has to follow to install and run it.
9. Create a demo video (minimum 5 minutes), demonstrating the installation details and example runs.

Note: There are no specific templates for writing the report. You can use your own template or write it in free format, but your report should address the items mentioned above in the same order they are given.

Grading:

Your result depends on the success of your work, your level of active engagement and above all, ability to explain what you learnt and how you implemented it.

Requirements for a pass grade (grade G):

1. Source dataset is curated, used directly by importing without any preprocessing
2. Solution is expected to have basic/simple generative capabilities or features only using the generative models from the course book.
3. Solution uses simple prompts to communicate with your app.
4. Solution runs from command line or notebook

Requirements for a higher grade (grade VG):

1. Source dataset is raw and needs preprocessing such as removal of unnecessary information, feature encoding, labeling, etc.
2. Solution is expected to have complex generative capabilities, models used could be from elsewhere outside the course book (The student identified and learnt on her/his own).
3. Solution uses relatively complex prompts (e.g. interactive with memory/conversation history) and prompts are refined to improve output quality and consistency.
4. Solution has a friendly interface such as a browser to interact with user (e.g. build a Streamlit app).

Note: In some cases, a subjective evaluation may be needed to set grades. For example, a student/group aimed for grade **VG** but the implementations are only partially complete, a **G** grade may be assigned.

Project ideas:

It would be great if you have already in mind something you would like to implement using generative AI (there are always fascinating or exciting ideas in this field!). You can also search online for possible applications that you can build with in the course timeframe.

Example project ideas (you may get inspiration from the ones below):

1. Conversational Chatbot (e.g. in healthcare or other domain specific areas)
2. Text-to-image art generator
3. Audio/music app, generates melody and lyrics
4. AI voice assistant for home use
5. AI document summarizer assistant for medical practitioners (e.g. clinical notes)
6. Image captioning app (take an image as input and generate caption/description of the image)